

BTC Market Intelligence Engine — Strategy Research Lab

Subsystem brief v2

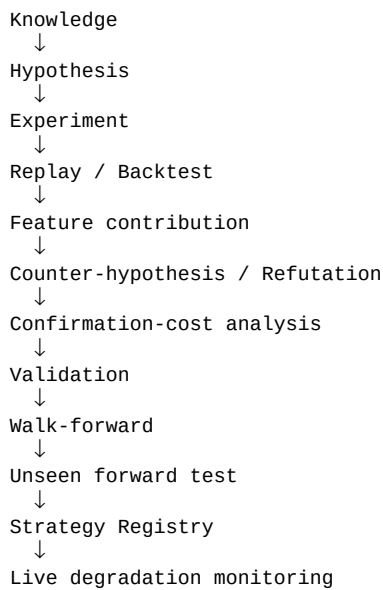
Architectural role: Application/research layer around the domain core

Purpose

Discover, formalize, test, validate and monitor candidate strategies from measured market behavior.

The lab is not allowed to rewrite the domain's numerical truth.

Research loop



Experiment definition

Every hypothesis should specify:

- Market State filters
- Regime filters
- Location
- Trigger
- Entry
- Invalidation
- Target / management
- fees
- slippage
- funding
- latency assumptions
- sample period
- data version
- feature version

Results are immutable and carry provenance.

Order-Flow Feature Contribution

Order-flow variables are first-class research features.

Experiments should be able to compare:

Base:
Location + Failed Auction

Add:
+ aggressive buy/sell imbalance
+ delta/CVD
+ absorption
+ effort/result
+ passive liquidity response

Measure whether these features improve expectancy, robustness and entry quality.

Do not assume that a visually convincing Bookmap pattern has predictive value.

Edge Contribution

Test a base hypothesis against incremental additions.

Example:

Base:
VAH + failed auction

Test additions:
+ CVD divergence
+ OI condition
+ ATR regime
+ absorption
+ effort/result
+ SFP

Measure:

- sample size
- expectancy
- profit factor
- win rate
- MFE / MAE
- drawdown
- entry displacement
- time-to-target
- robustness by regime

More confirmation is not automatically better.

Confirmation cost

Compare:

- early trigger
- trigger + one confirmation
- trigger + multiple confirmations
- candle-close confirmation

Account for worse entry price, missed trades and changed R/R.

Agent / LLM role

Agents may:

- generate hypotheses
- plan experiments
- combine candidate features
- analyze results
- challenge/refute assumptions
- prioritize research

Agents may not become the authoritative source for numerical facts or performance metrics.

Hexagonal architecture

The lab calls application/domain ports such as:

- ``HistoricalDataProvider``
- ``BacktestEngine``
- ``StrategyRepository``
- ``ResearchResultStore``

Concrete engines can be replaced without changing the research-domain rules.

Anti-overfitting

Require:

- provenance
- train/validation separation
- walk-forward validation
- unseen forward testing
- minimum sample sizes
- leakage controls
- versioned features and data